

Class 12: Wi-Fi Sensing

Where Is the Phone? From Signal Strength to Angle and Distance

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CIS 4930/5930 Future Edge Networks

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Where is the phone?

Applications:

- Navigation: both outdoors (GPS) and indoors (e.g., inside museum)
- Location based services: Tagging, Reminder, Ads
- Virtual Reality and Motion Capture
- Gestures, writing in the air
- Behavioral Analytics (Health, activities, etc.)
- Locating misplaced items (keys)
- Security (e.g., only want to give WiFi access to customers inside a store)
- Delivery drones



Every one of these needs a position — **indoors**, where the one positioning system everybody knows does not reach.

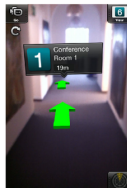
Figure: Adib & Eid, MIT MAS.S60 Lecture 2, "Fundamentals of Wireless Sensing & Localization"

Outdoors: GPS. Indoors: nothing good

- Outdoors: **GPS**
 - **Accurate for navigation (meters)**
 - **Signals fade in indoor environments**



- Precise and rapid indoor location enables:
 - Augmented reality on the smartphone, wearable or glasses
 - Fine-grained location in supermarkets, libraries or museums
 - Controlling network access based on desk or room
- Known technologies: **not accurate enough (WiFi)**, **require dedicated infrastructure (ultrasound)** or **require cameras and good light conditions (vision)**



Every building already has Wi-Fi.
Can the access point itself work out where the phone is?

Figure: Xiong & Jamieson, ArrayTrack talk, NSDI 2013

Who does the locating?

- Device based: A device uses incoming signal from one or more “anchors” to determine its own location



- Example: GPS

- Network based: Anchors (or Access points) use the signal coming from device to determine its location



- Example: Radar

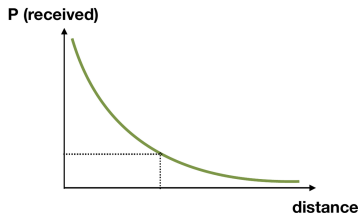
Today: **network-based**. The AP hears the phone's packet anyway. What can it measure from it?

signal strength → phase → angle → distance → both

Figure: MIT MAS.S60 Lecture 2

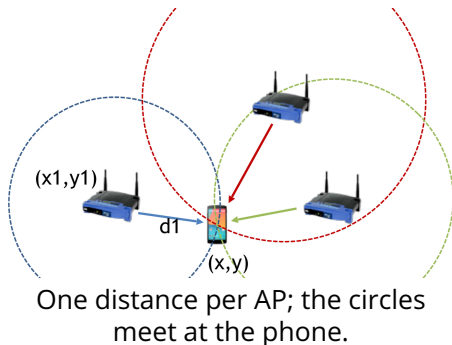
Attempt 1: louder means closer

Power is proportional to $1/d^2$



Power $\propto 1/d^2$: read a power, get a distance.

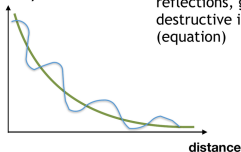
Figures: MIT MAS.S60 Lecture 2



Why RSSI stops at room level

Power is proportional to $1/d^2$

P (received)

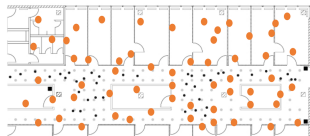


Con 2: Multipath: Due to reflections, get constructive and destructive interference (equation)

Reflections add and cancel: power wiggles with position.

Solution: Fingerprinting

i.e., measuring device records signal strength fingerprints at each location



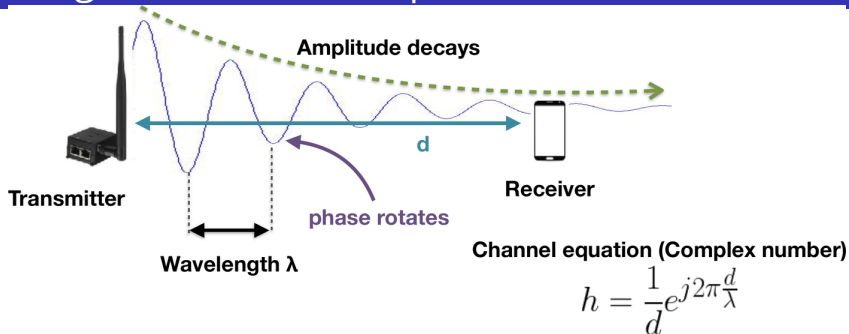
Workaround: record a fingerprint at every dot.

One number per packet, summed over every reflection:
room-level.

For a desk-level answer we need a finer ruler.

Figures: MIT MAS.S60 Lecture 2

The signal is a wave; the phase is a ruler

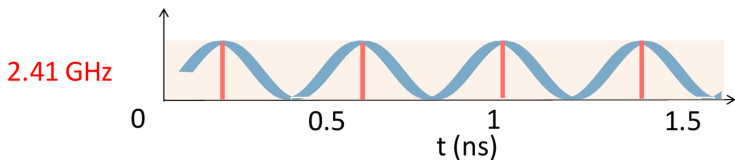


- Amplitude decays — that was RSSI. **Phase turns 360° every wavelength.**
- 5 GHz: $\lambda = 6$ cm, so 1° of phase ≈ 0.17 mm of path.
- The card already computes h per subcarrier, per antenna, to decode (Class 8): that is **CSI**.

CSI = Channel State Information: the h values the receiver estimated from the packet preamble

Figure: MIT MAS.S60 Lecture 2

...but the ruler has no numbers on it



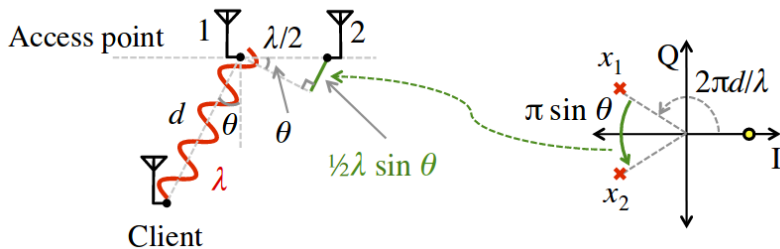
$$\phi = 2\pi f t \bmod 2\pi$$

Every red tick looks identical to the receiver. A 5 m path has turned ≈ 87 times; you see only the leftover fraction.

You cannot read one phase. You can read the **difference** between two:
two **antennas** $\rightarrow$ **angle**; two **frequencies** $\rightarrow$ **distance**.

Figure: Vasisht, Kumar & Katabi, Chronos talk, NSDI 2016

Two antennas hear the wave at two phases



Antenna 2 is $\frac{\lambda}{2} \sin \theta$ farther, so it lags by $\pi \sin \theta$:

$$\theta = \arcsin\left(\frac{\angle x_2 - \angle x_1}{\pi}\right)$$

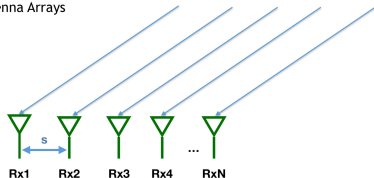
Phone at 30° : a **quarter turn** per antenna.

Spacing is $\lambda/2$ so the step itself never wraps.

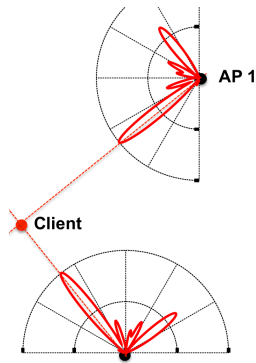
Figure: ArrayTrack, NSDI 2013 (as redrawn in Princeton COS463 Lab 5)

More antennas: a spectrum of power vs. bearing

Use Antenna Arrays



N antennas, $N - 1$ phase steps:
sweep θ in software.

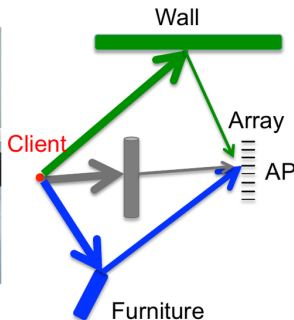
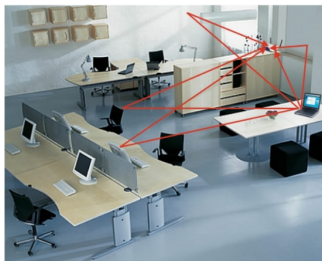


A peak in the spectrum is a bearing.
Two APs' bearings cross at the phone.

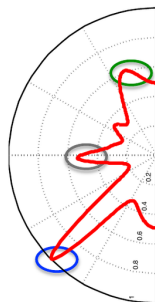
Figures: MIT MAS.S60 Lecture 2 (left); ArrayTrack talk, NSDI 2013 (right)

The office fights back: multipath

- **Problem #1:** Strong multipath reflections indoors
- **Problem #2:** Direct path attenuated or completely blocked
 - Direct path signal may not be the strongest



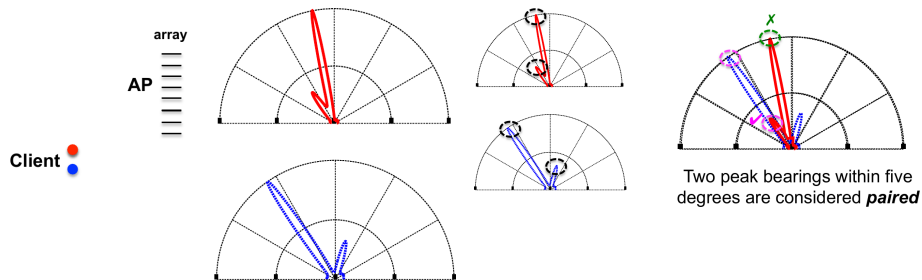
AoA spectrum



Three lobes in the spectrum. Which one is the phone?

Figure: ArrayTrack talk, NSDI 2013

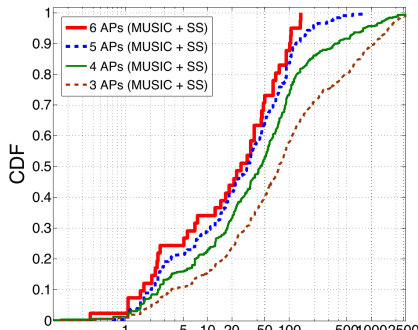
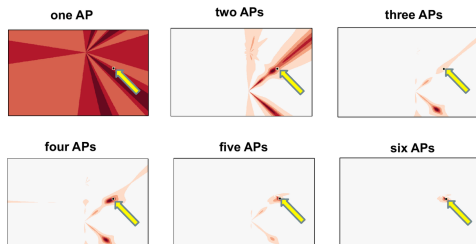
ArrayTrack's trick: the direct path holds still



Move the client a few cm. Reflected bearings swing; the direct bearing stays. Keep the peaks that reappear within 5° .

Figures: ArrayTrack talk, NSDI 2013

More APs, sharper dot: 23 cm



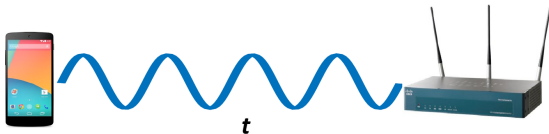
One AP gives a bearing, not a point. Six 8-antenna APs: **23 cm median**.

Angle alone needs several APs around the room.
Your home has one.

Figures: ArrayTrack talk, NSDI 2013

Homes have ONE access point

Homes and small businesses have **ONE** access point (AP)

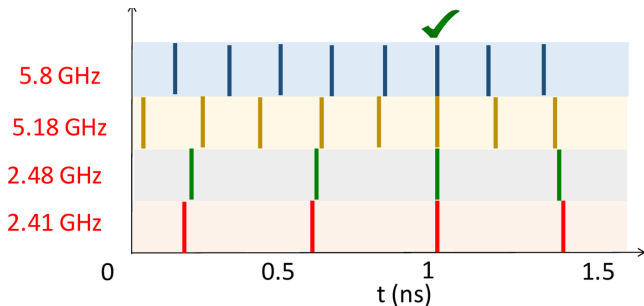


$$\text{Phase of the signal}(\phi) = 2\pi f t \bmod 2\pi$$

One AP needs angle and distance. Distance = $c \times$ propagation delay, and the delay sits in the phase — wrapped.

Figures: Chronos talk, NSDI 2016

Many frequencies agree on only one delay

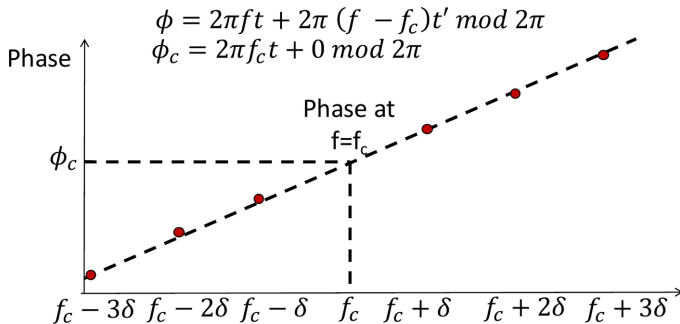


Each frequency wraps at its own period. Only the true delay is consistent with all of them.

Same sentence as last section: a **difference across a small step** beats wrapping — now the step is in frequency, not space.

Figure: Chronos talk, NSDI 2016

OFDM hands you the frequencies for free

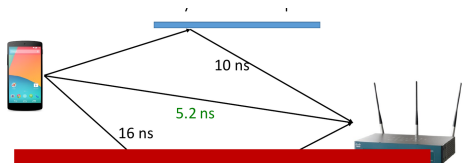


Phase vs. subcarrier is a **straight line**; its slope is the delay. 5 m path:
 $\approx 120^\circ$ of fall across one 20 MHz packet.

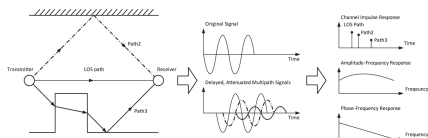
You have known this since Class 8: every packet is 52+ equally spaced tones, each reporting its own phase.

Figure: Chronos talk, NSDI 2016

Multipath: the shortest delay is the distance



Distance to source corresponds to the smallest delay.



IFFT across subcarriers: one spike per path.

A reflection always travels farther. $5.2 \text{ ns} \times c = 1.6 \text{ m}$ to the phone.

Figures: Chronos talk, NSDI 2016 (left); Yang et al., Tsinghua "Hands-on Wireless Sensing" tutorial (right)

The catch: the receiver's own delay



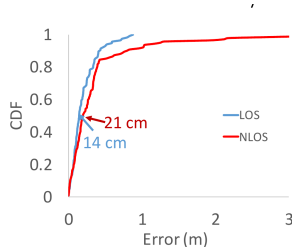
Distance = speed of light $\times$ propagation delay

Measured delay = propagation delay + detection delay

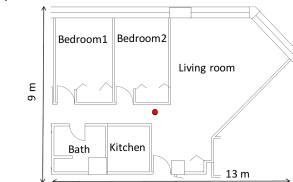
- Detection delay is larger than propagation delay and changes every packet.
- Chronos: the phase at the carrier is immune — read it off the OFDM line; a packet/ACK pair cancels clock offsets.
- Resolution is bandwidth: 20 MHz separates paths only ~ 15 m apart, so Chronos hops across all Wi-Fi bands.

Figure: Chronos talk, NSDI 2016

What a single access point achieves



Distance: **14 cm** (LOS),
21 cm (NLOS)



Right room **94%** of the
time

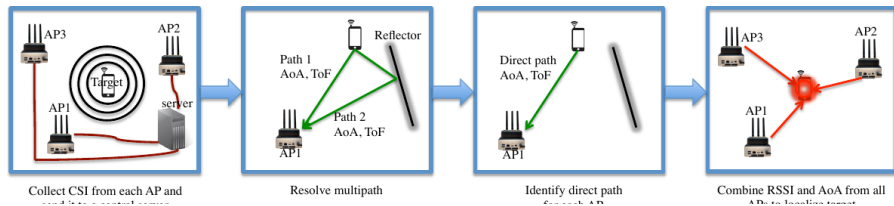


Drone follows at 1.4
m, **4 cm** error

One off-the-shelf Wi-Fi card. No infrastructure beyond the AP.

Figures: Chronos talk, NSDI 2016

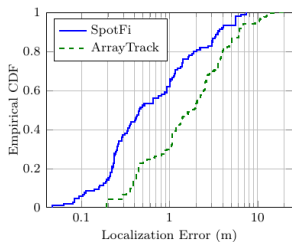
Angle and delay from the same packet



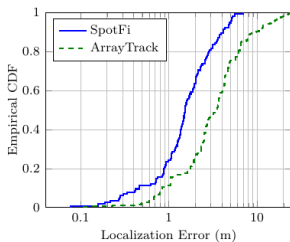
- 3 antennas $\times$ 30 subcarriers = **90 sensors** in one packet's CSI.
- Every path becomes an (**angle**, **delay**) pair; fit them jointly.
- Direct path = the **earliest** pair with a stable bearing. Then cross the bearings.

Figure 1: Kotaru, Joshi, Bharadia & Katti, SpotFi, SIGCOMM 2015

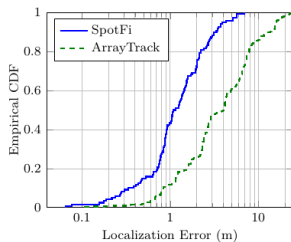
A stock 3-antenna AP: 40 cm median



(a) Indoor office deployment



(b) High NLoS deployment



(c) Corridors

Same office, same 3-antenna hardware: SpotFi **0.4 m** median vs. 1.8 m for angle-only. Without line of sight: 1.6 m.

“Where is the phone?” — answered to 40 cm by access points that were already on the wall.

Figure 7: SpotFi, SIGCOMM 2015

The ladder we climbed

Read...	across...	gives	system
signal strength	—	a room	fingerprinting
one phase	—	nothing (wrapped)	—
phase difference	antennas	angle	ArrayTrack: 23 cm, 6 APs
phase difference	subcarriers	distance	Chronos: 14 cm, 1 AP
both	both	every path	SpotFi: 40 cm, stock APs

Phase is a 6 cm ruler with no absolute marks.
Everything above RSSI is a **difference** of phases along some axis.

Next: the same signal sees the room



Seeing people through a wall with Wi-Fi

- **Class 13:** 2FiA reads breathing and heartbeat out of CSI.
- **Class 14:** the AP already broadcasts compressed CSI — beamforming feedback.

Try it (free, with code):

- Tsinghua [Hands-on Wireless Sensing](#):
`naive_aoa`, `naive_tof`
- Princeton COS463 [AoA / MUSIC lab](#) in NumPy

Your phone never opted in. Who may run a ruler made of your Wi-Fi?